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SRI RAMAKRISHNA ENGINEERING COLLEGE, COIMBATORE

(Autonomous Institution, Reaccredited by NAAC with 'A+' Grade Approved by AICTE and Permanently affiliated to Anna University, Chennai)

AUTONOMOUS EXAMINATIONS - NOV'2024

B.E – COMPUTER SCIENCE AND ENGINEERING

20CS2E22 - DIGITAL FORENSICS

Answer ALL questions

Duration: 3 Hours

Maximum: 100 Marks

PART - A

(7x 1 = 7 Marks)

1. **CO1** Which technique is most commonly used to recover erased or damaged data in digital forensics?
 - a) Data carving
 - b) Disk wiping
 - c) Cloud backup retrieval
 - d) File compression
2. **CO1** What is tracked during network traffic analysis?
 - a) System temperature
 - b) Data flows and communication patterns
 - c) Hardware device types
 - d) Memory leaks
3. **CO2** Which of the following is analyzed during browser forensics to uncover web activity?
 - a) Operating system logs
 - b) Cache and temporary internet files
 - c) Hardware memory
 - d) Email communication
4. **CO2** Which of the following is a common internet threat investigated in digital forensics?
 - a) DDoS attacks
 - b) Phishing attacks
 - c) Data encryption
 - d) Web search algorithms
5. **CO3** Which tool is commonly used for reverse engineering malware?
 - a) Wireshark
 - b) OllyDbg
 - c) Nmap
 - d) ClamAV

6. **CO2** Which of the following types of files can provide useful information when you are examining an e-mail server?
 a) .dbf files b) .emx files c) .log files d).slf files
7. **CO3** What is the section of IPC for cyber crime?
 a) Indian Penal Code, 1890 b) Indian Penal Code, 1982
 c) Indian Penal Code, 1980 d) Indian Penal Code, 1990

Fill in the blanks: **(3x 1 = 3 Marks)**

8. **CO2** _____ techniques can be used for surveillance which can record the passwords, username, and confidential information.
9. **CO1** _____ can show the pattern of transmitting data to or from a particular IP address
10. **CO3** _____ categories of information is stored on a SIM card.

PART - B **(5x 3 = 15 Marks)**

11. **CO1** Name a few common forensic software tools and their purpose in digital investigations.
12. **CO2** How is messenger forensics conducted for platforms like AOL, Yahoo, MSN, and chat services?
13. **CO2** What should be included in a forensic report outlining the findings of a digital investigation?
14. **CO3** How does reverse engineering help in malware analysis, and what tools are commonly used?
15. **CO3** List out the techniques are used to discover and extract malware from a Windows system?

PART - C **(5 x 15 = 75 Marks)**

Q.No: 16 Compulsory Question

Q.No: 17 to 22 Answer any FOUR Questions

Compulsory Question:

16. **CO1** i) Explain the Concept of Order of Volatility and its Role in Digital and Network Forensics. (8)

- CO1** ii) Enumerate how do Automated Search Techniques Assist in Digital and Network Forensics Investigations? (7)
17. **CO1** Summarize the standard procedures in Digital and Network Forensics, including incident verification, system identification, data recovery, and disk imaging. (15)
18. **CO2** i) Interpret the role of the Information Technology Act in digital forensics. How does this legislation govern the collection, analysis, and presentation of digital evidence in a court of law? (8)
- CO2** ii) Describe the process of collecting and analyzing data from mobile devices such as cell phones, PDAs, Blackberries, iPhones, iPods, and iPads. What are the challenges and methods used to extract forensic evidence from these devices? (7)
19. **CO2** Classify the various internet and World Wide Web threats such as email threats, chat room dangers, hacking, and illegal access. How can these threats be investigated and mitigated in digital forensics? (15)
20. **CO3** i) Assess the techniques and tools used to analyze a live Windows system for malware. How can malware be identified and mitigated while the system is running? (8)
- CO3** ii) Outline the methods for discovering and extracting malware from Linux systems. How do these methods differ from Windows systems, and what unique tools are used in the process? (7)
21. **CO3** Compare and contrast the methods used for analyzing malware on live Linux systems with those on Windows systems. What unique challenges and solutions exist for Linux-based malware analysis? (15)

22. **CO2** Assess the key techniques involved in browser forensics, particularly in analyzing cache, temporary internet files, cookies, and reconstructing web browsing activity? Provide examples of how these methods can be applied in a forensic investigation. (15)
